

RobustDRNet: A clinically-aligned hybrid ensemble model with multi-method explainability for lesion-aware diabetic retinopathy grading

Pir Bakhsh Khokhar ^{1,*}, Viviana Pentangelo , Carmine Gravino , Fabio Palomba 

Department of Informatics, University of Salerno, Via Giovanni Paolo II, 132, Fisciano, 84084, Salerno, Italy

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ABSTRACT

Diabetic retinopathy (DR) screening requires artificial intelligence (AI) models that are not only highly accurate in grading five clinical stages but are also capable of generating quantitatively evaluated lesion-aware explanations to earn the trust of clinicians. We propose *RobustDRNet*, a hybrid ensemble model that combines local convolutional features from Residual Network-34 (ResNet-34) and ConvNeXt-Tiny with global transformer embeddings from Vision Transformer Base/16 (ViT-B16) via two-stage feature fusion and a disentangled multilayer perceptron (MLP), followed by a logistic regression stacking meta-learner for prediction aggregation. To address severe class imbalance, our training pipeline employs stratified sampling, contrast-limited adaptive histogram equalization (CLAHE) for contrast enhancement, strong data augmentation, and class-weighted focal loss. Evaluated on the Asia Pacific Tele-Ophthalmology Society (APTOS) 2019 dataset, RobustDRNet achieved 88.4% validation accuracy, a 0.967 macro-averaged area under the receiver operating characteristic curve (macro-AUC), and Cohen's kappa of 0.823, outperforming individual backbones and simple voting ensembles. In addition to classification performance, we integrated six complementary explainable AI (XAI) techniques: Gradient-weighted Class Activation Mapping ++ (Grad-CAM ++), Integrated Gradients, attention rollout, SHapley Additive exPlanations (SHAP), Local Interpretable Model-Agnostic Explanations (LIME), and Testing with Concept Activation Vectors (TCAV). Each technique was quantitatively benchmarked against expert-annotated lesion maps from the Indian Diabetic Retinopathy Image Dataset (IDRiD). Saliency maps achieved mean Intersection over Union (IoU) scores of 0.06 for Grad-CAM ++ and approximately 0.10 for Integrated Gradients; SHapley Additive exPlanations (SHAP) perturbations showed a deletion drop of 0.25 and an insertion gain of 0.22; and TCAV achieved complete classifier-level TCAV alignment (score = 1.0) with clinically coherent, grade-wise importance trajectories. By combining competitive grading performance with multi-perspective and quantitatively evaluated interpretability, RobustDRNet provides a promising DR screening framework whose decisions are supported by lesion-aware explanatory evidence.

1. Introduction

Diabetic retinopathy (DR) is one of the main complications of diabetes mellitus (DM), which affects more than one-third of diabetic patients worldwide and demonstrates noteworthy disparities between regions in terms of social standing regarding disease occurrence and care availability. According to the International Diabetes Federation (IDF), diabetes affected 537 million adults worldwide in 2021, comprising 9.3% of the global population, with projections indicating that 783 million adults will be affected by 2045 (Federation, 2021). DR screening

should be performed for every patient with diabetes because this disease occurs in 94% of diabetics throughout their lives, and 28 million patients face sight-threatening development of either proliferative diabetic retinopathy (PDR) or diabetic macular edema (DME) (Wang et al., 2022). Vision-threatening complications of the disease affect populations in low- and middle-income nations most severely because poor glycemic control, hypertension, and delayed screening contribute to faster disease progression.

While early detection is crucial for avoiding irreversible vision loss, the standard diagnostic workflow is the manual grading of fundus im-

* Corresponding author.

E-mail addresses: p.khokhar@studenti.unisa.it (P.B. Khokhar), vpentangelo@unisa.it (V. Pentangelo), gravino@unisa.it (C. Gravino), fpalomba@unisa.it (F. Palomba).

URL: <https://rubrica.unisa.it/persona?matricola=066182> (P.B. Khokhar), <https://vipenti.github.io/> (V. Pentangelo), <https://docenti.unisa.it/004724/home> (C. Gravino), <https://docenti.unisa.it/027888/home> (F. Palomba)

¹ This is the first author footnote, also applicable to the third author.

ages by ophthalmologists, which is a time-consuming and laborious process that is susceptible to inter-observer variability, particularly in the early and borderline stages (Krause et al., 2018). These limitations have led to an increased interest in the development of advanced AI systems capable of automating DR grading and achieving expert-level performance and consistency. In particular, CNNs have been widely adopted for automated DR classification owing to their ability to extract localized features from fundus images (Gargeya & Leng, 2017; Gulshan et al., 2016; He et al., 2016; Huang et al., 2017; Tan & Le, 2019). To address the limitations of CNNs concerning their narrow receptive fields and struggles with imbalanced DR datasets (Quellec et al., 2017), Vision Transformers (ViTs) have emerged as an alternative, leveraging self-attention to model global spatial dependencies (Dosovitskiy et al., 2020). Moreover, applications to DR grading (Wu et al., 2021) have shown higher sensitivity to spatially dispersed lesions.

Despite significant progress in leveraging CNNs and ViTs for DR classification, current approaches still have critical limitations. CNNs and ViTs are typically used in isolation, each capturing only partial aspects of retinal pathology, local or global, without integrating their complementary strengths. Moreover, explainability remains underdeveloped: most studies rely solely on Grad-CAM and lack frameworks that combine pixel- and concept-level interpretations. When used, XAI techniques are rarely subjected to quantitative evaluation, leaving open questions regarding their robustness, fidelity, and alignment with clinical reasoning. Finally, many DR pipelines fail to address severe-stage class imbalances, often overlooking tailored strategies, such as adaptive loss functions or label smoothing, which are essential to ensure consistent performance across all disease stages.

To address these limitations, we introduce RobustDRNet, a unified ensemble combining ViT-B/16's global attention (Dosovitskiy et al., 2020) with ResNet-34 (He et al., 2016; Wu et al., 2021) and ConvNeXt's local feature extraction (Liu et al., 2022), along with a customized pipeline that addresses dataset imbalances through class-weighted loss, label smoothing, and enhanced augmentations. This hybrid deep learning framework achieved superior performance across all five DR stages, offering an integrated solution for DR classification. RobustDRNet integrates six XAI methods (Grad-CAM + +, SHAP, LIME, Integrated Gradients, Attention Rollout, TCAV) to generate both pixel-level and concept-level insights. The model's interpretability is rigorously validated using quantitative benchmarks, including localization fidelity (w.r.t. lesion masks), perturbation stability, inferential faithfulness, and TCAV concept alignment, ensuring trust and transparency in its predictions.

The remainder of this paper is organized as follows: Section 2 presents a theoretical background and reviews related work on deep learning for DR classification and explainability in medical AI. Section 3 describes the model architecture, training strategies, and explainability integration. Section 4 outlines the experimental setup. Section 5 reports performance and interpretability results. Section 6 discusses key findings and implications. Section 7 concludes the work and outlines future directions.

2. Background & related work

This section introduces the clinical and technical background of our study, reviews the current state of the art in deep learning for diabetic retinopathy classification, and discusses related work with a focus on the key limitations that our approach aims to address.

2.1. Background

DR is a progressive microvascular complication of diabetes mellitus that affects the retinal blood vessels due to prolonged hyperglycemia, oxidative stress, and ischemic damage. DR manifests through specific lesion types such as microaneurysms, intraretinal hemorrhages, hard exudates, and cotton wool spots, eventually leading to neovascularization in the proliferative stage (PDR). Clinically, DR is categorized into

five stages, ranging from no DR to mild, moderate, and severe non-proliferative DR (NPDR), and finally proliferative DR (PDR), which is associated with high risks of vitreous hemorrhage and retinal detachment. Timely detection of early-stage lesions is critical to prevent irreversible vision loss. However, manual grading, the standard diagnostic approach, is time-consuming and susceptible to inter-observer variability (Stitt et al., 2016).

CNNs have been widely applied to DR detection due to their ability to learn hierarchical local patterns from fundus images. These models leverage convolutional filters to extract spatial features that correspond to lesion-level abnormalities such as microaneurysms and exudates. Early works, including (Gulshan et al., 2016), demonstrated expert-level performance in referable DR detection, while Gargeya and Leng (2017) showed high sensitivity using handcrafted preprocessing and CNNs. Subsequent architectures such as ResNet (He et al., 2016), DenseNet (Huang et al., 2017), and EfficientNet (Tan & Le, 2019) have further improved classification performance through increased depth, residual connections, and parameter efficiency. Nevertheless, CNNs are inherently limited by their localized receptive fields, which makes it challenging to capture global context across the retina, particularly for distinguishing intermediate DR stages. Furthermore, they often underperform in imbalanced datasets, where severe DR classes are underrepresented (Quellec et al., 2017).

Vision Transformers (ViTs) have emerged as an alternative to CNNs for image classification tasks, including DR grading. Unlike CNNs, ViTs utilize self-attention mechanisms to model long-range dependencies between image patches, enabling the network to capture global structural information. The standard ViT architecture introduced by Dosovitskiy et al. (2020) has shown competitive or superior performance to CNNs given sufficient pretraining data. In the context of DR grading, Vision Transformer-based models have shown that self-attention can capture global retinal structure and spatially distributed lesion patterns from fundus images (Mohan et al., 2022; Wu et al., 2021). However, ViTs typically require large-scale training datasets to converge effectively, and their lack of inductive biases (e.g., locality, translation invariance) makes them less suited to fine-grained lesion detection without explicit architectural enhancements or augmentations. Moreover, attention maps produced by transformers may not consistently align with clinically meaningful regions, limiting their interpretability.

2.2. Related work

Deep learning models have advanced DR detection and grading through multiple stages, progressing from CNN-based models to transformer-based and hybrid approaches. The analysis of XAI has emerged parallel to how systems interpret data, because XAI matters for clinical adoption through its transparent lesion-aware explanatory reasoning along with accurate classification.

Many works have focused on CNN-based models for DR detection, often prioritizing classification accuracy without incorporating interpretability mechanisms. Goh et al. (2024) proposed a pipeline combining VGG-16 with an SVM for binary classification, achieving strong performance but offering no insight into feature-level reasoning. Similarly, Yasashvini et al. (2022) designed a sequential CNN for five-class DR grading using bilateral and Gaussian filtering, yet their approach lacked any form of explanation or attention-based component. Other studies such as (Bilal et al., 2021), (Wahab Sait, 2023), and (Bhoopal et al., 2024) introduced lightweight CNN architectures with various preprocessing strategies, including contrast enhancement and wavelet transforms, but none of these integrated any interpretability tools. Hybrid methods using CNNs with traditional classifiers like SVM and k-NN, as in Ishtiaq et al. (2023), also omitted explainability and were restricted to binary settings.

Gargeya and Leng (2017) focused on telemedicine applications with a custom CNN but did not consider stage-wise classification or explanatory outputs. Finally, Anoop et al. (2022) developed a dual CNN archi-

Table 1
Qualitative comparison of recent hybrid DR models with RobustDRNet.

Studies	Architecture	Ensemble	Explainability
Khan et al. (2023)	CNN-Transformer	No	Grad-CAM (qualitative)
Wu et al. (2021)	Transformer-centric	No	Attention maps (qualitative)
Mohan et al. (2022)	CNN-Transformer	No	Attention maps (qualitative)
RobustDRNet	CNN + ConvNeXt-Tiny + ViT-B/16	Yes	Multi-method, quantitative

texture for binary DR detection, but failed to address class imbalance or interpretability. These contributions demonstrate the effectiveness of CNNs in DR detection; however, they remain limited by their lack of architectural complementarity and interpretability, leaving large room for improvement in supporting clinical decision processes.

To bridge the gap between convolutional and transformer-based approaches, several studies have proposed hybrid CNN-Transformer architectures for diabetic retinopathy (DR) grading. Notable examples include DR-CCTNet ([Khan et al., 2023](#)), which integrates convolutional feature extraction with transformer attention and employs Grad-CAM for visual explanation, as well as ([Wu et al., 2021](#)) and ([Mohan et al., 2022](#)), which leverage self-attention mechanisms to enhance global context modeling in retinal images. These hybrid models demonstrate improved sensitivity to spatially distributed lesion patterns compared to standalone CNNs or vision transformers, as summarized in [Table 1](#).

However, existing hybrid approaches rely on a single hybrid backbone, limiting representational diversity and restricting error compensation across heterogeneous feature spaces. In addition, interpretability in these models is primarily qualitative and rarely validated quantitatively at the lesion level. These limitations motivate ensemble-based hybrid frameworks that combine multiple architectural inductive biases while supporting systematic, quantitative explainability.

From a comparative perspective, machine learning (ML) and deep learning (DL) approaches exhibit complementary strengths in diabetes and diabetic retinopathy (DR) prediction.

! Limitation 1

Existing DR models are either CNN-only (local focus) or transformer-only (global focus); none of them combine both architectures in a unified ensemble.

Recent advances in artificial intelligence for diabetic retinopathy (DR) grading extend beyond backbone selection and include improvements in preprocessing, imbalance-aware training, and clinically oriented interpretation. In image-based DR analysis, contrast enhancement and normalization (e.g., CLAHE-based pipelines) are commonly applied to improve lesion visibility and reduce acquisition variability ([Hayati et al., 2023](#)). Class imbalance, which is prevalent in DR datasets, is addressed through imbalance-aware sampling and cost-sensitive learning strategies to improve sensitivity for underrepresented disease stages ([Queltec et al., 2017](#)). In parallel, transformer-based and hybrid architectures enhance global context modeling and improve sensitivity to spatially dispersed lesion patterns ([Mohan et al., 2022](#); [Wu et al., 2021](#)). Collectively, these developments highlight a shift toward clinically aligned AI pipelines; however, many approaches still address these components in isolation, motivating unified frameworks that integrate robust multi-stage grading with systematically evaluated explainability.

To address the lack of interpretability in CNN-based DR models, a number of works have integrated visual explanation techniques, primarily relying on saliency maps. [Khan et al. \(2023\)](#) introduced DR-CCTNet, a compact convolutional transformer that uses Grad-CAM to highlight relevant retinal lesions across multiple datasets, though the explanations remained qualitative. [Hayati et al. \(2023\)](#) proposed a min-

imal CNN for five-class DR classification using CLAHE enhanced inputs and Grad-CAM++ to visualize attention regions, yet without leveraging standard attribution metrics or validating the explanations at lesion level. In a similar vein, extended Grad-CAM by pairing it with natural language explanations using clinical terminology, aiming for improved semantic interpretability, though they did not include any performance or fidelity metrics. These studies predominantly focus on pixel-level heatmaps. Building upon their results, these findings motivate the incorporation of higher-level conceptual reasoning and systematic evaluation to enhance the clinical trustworthiness and interpretive depth of DR prediction models.

! Limitation 2

Most explainability studies rely solely on Grad-CAM, lacking a unified framework that merges pixel-level heatmaps with concept-level interpretations.

! Limitation 3

XAI techniques in DR research have rarely been evaluated quantitatively; localization accuracy, robustness to perturbations, causal consistency, and TCAV alignment have not been systematically measured.

Recent efforts have explored hybrid and transformer-based architectures while integrating more advanced explainability mechanisms. [Kim et al. \(2018\)](#) introduced TCAV to assess concept relevance in CNN outputs, focusing on lesion types such as microaneurysms and exudates. [Abdelmaksoud et al. \(2021\)](#) introduced an attention-based CNN for spatial lesion segmentation using attention maps. Ensemble models like ([Shorfuzzaman et al., 2021](#)) used SHAP for feature-level explanations. [Yasashvini et al. \(2022\)](#) used CNN and hybrid CNN models for DR classification, while [Wu et al. \(2021\)](#) explored ViT based DR grading; however, neither provided multi-method lesion-level explainability. [Kadry et al. \(2023\)](#) merged CNNs with handcrafted features in an ensemble focused on clinical feasibility but excluded XAI. [Dai et al. \(2021\)](#) developed a lesion-level CAD system using Faster R-CNN and ResNet-101 for detecting microaneurysms and hemorrhages, though it lacked grading and explanation capabilities. While these approaches combine strong architectures with interpretability, class imbalance challenges remain and mitigation strategies are applied rarely.

! Limitation 4

DR grading pipelines often ignore severe-stage class imbalances and fail to use adaptive weighting, label smoothing, or augmentation to ensure uniform performance across all grades.

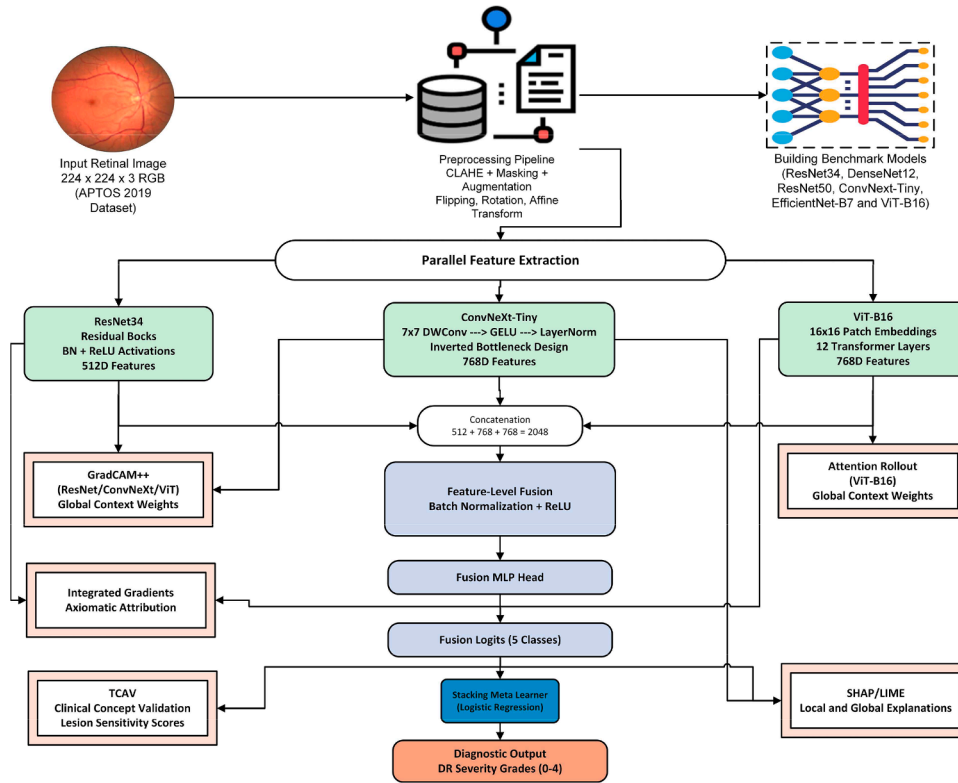


Fig. 1. RobustDRNet architecture. Preprocessed fundus images are encoded by ResNet-34, ConvNeXt-Tiny, and ViT-B/16; their embeddings are concatenated and passed through a fusion MLP to produce five-class fusion logits, which are input to a logistic-regression stacking meta-learner for final DR grade prediction (0–4). Post-hoc explainability methods are applied separately and do not affect prediction.

3. Proposed RobustDRNet model for diabetic retinopathy grading

3.1. Overview of proposed methodology

We propose a robust and explainable deep-learning framework for automatic DR severity classification based on retinal fundus images. Our method provides high-performance deep learning models with clinically meaningful interpretability to address the urgent need for early and accurate diagnosis. We propose an architecture that uses a unified ensemble of CNNs and ViTs to encode fine-grained lesion-level features as well as broader retinal patterns.

ResNet-34 and ConvNeXt-Tiny have localized pathological sign detection abilities in microaneurysms and hemorrhages, whereas ViT-B/16 is good at a long range and captures the global retinal context. The framework improves classification accuracy over all five levels of DR severity by fusing these complementary feature representations. We included a complete explainability suite to ensure transparency and added clinical trust to the model decisions by providing spatial and concept-based interpretations.

We organize the proposed methodology into six main stages: (i) image preprocessing and augmentation, (ii) building benchmark models, (iii) deep feature extraction with multiple backbones, (iv) feature level fusion, (v) ensemble-based classification, and (vi) explainability with global and local interpretation techniques. Fig. 1 presents a top-level overview of the entire pipeline, which shows the progression of the raw image input to the final diagnostic output. This design is suitable for deployment in real-world clinical DR screening applications because it is an integrated design that facilitates both high-performance and interpretability.

3.2. Image preprocessing and augmentation

In medical imaging, the high quality and consistency of the input are critical for reliable model performance. For this purpose, we designed a structured preprocessing and augmentation pipeline customized to the properties of the APTOS 2019 Blindness Detection retinal fundus images dataset (Asia Pacific Tele-Ophthalmology Society, 2019; Porwal et al., 2018).

It is used to standardize, enhance, and enrich input images for accurate detection of DR features. All images were first resized to a uniform $224 \times 224 \times 3$ pixel resolution to comply with the chosen backbones (ResNet-34, ConvNeXt-Tiny, and ViT-B/16) while balancing the computation efficiency and detail preservation. The variable resolution and acquisition conditions of the image necessitate resizing for consistent batch training and inference.

CLAHE was applied to increase the visibility of subtle pathological features, improving the local contrast by shifting the pixel intensities within small tiles without amplifying the noise. As these lesions represent the ground truth for early stage diseases such as microaneurysms and hemorrhages, this step is also particularly effective in this regard. To isolate the retinal area and remove non-informative backgrounds such as black borders, labels, and imaging artifacts, we also applied circular masking. We restricted the visual attention to diagnostically relevant signal content (masking the central retinal area) and prevented the learning of spurious correlations.

During training, we were able to address the dataset imbalance and improve the generalization using a range of data augmentation schemes. Finally, the simulated imaging orientation (random horizontal and vertical flipping) is varied. On the other hand, rotations are shifted by

± 15 and ± 30 to compensate for patient position change, and affine transformations, such as scaling and translation, are implemented to model geometric variability. Furthermore, the model encountered a dynamically changing dataset and probabilistically applied these augmentations. To this end, we utilized this strategy to efficiently increase the size of the training set, decrease overfitting, and create a more robust model for changes in pose, scale, and illumination.

Overall, our preprocessing and augmentation module transformed the raw fundus images into high-quality, standardized, and diverse inputs. This foundational task was critical for the model to discover fine-grained DR features and generalize to different clinical imaging conditions.

3.3. Building benchmark models

We built a series of benchmark models, each trained independently on the preprocessed dataset, to establish a good baseline and individual contribution of various deep-learning architectures. We considered widely adopted convolutional networks and transformer-based architectures found to be the best in the literature, including ResNet-34, DenseNet-121, ResNet-50, ConvNeXt-Tiny, EfficientNet B7, and ViT-B/16 (Dosovitskiy et al., 2020; He et al., 2016; Huang et al., 2017; Liu et al., 2022; Tan & Le, 2019). The goal of this stage was two-fold: (i) to measure the performance of each backbone under an identical training regime to justify the selection of the best performers as part of our final hybrid ensemble, and (ii) to select the top performing architectures for inclusion.

Softmax activation was used in the last layer for each model to output five probability scores for each of the five DR severity grades. To ensure a fair comparison, all benchmark models were trained using the same preprocessing pipeline and data augmentation strategies. We used the Adam optimizer with a learning rate of 1×10^{-4} and tracked the model performance with early stopping and validation loss. For class imbalance and to improve generalizability, training and evaluation were performed using a stratified 5-fold cross validation. Key evaluation metrics included accuracy, macro F1-score, Cohen's kappa, and AUC (Gargeya & Leng, 2017; Gulshan et al., 2016).

The results of the analysis of these cases helped identify the strengths and weaknesses of the sensitivity of each model to different DR severity levels.

Although all six models performed competitively, ResNet-34, ConvNeXt-Tiny, and ViT-B/16 appeared to have the best balance sensitivity, specificity, and generalization. Based on the above analysis, feature fusion and ensemble pipelines were constructed using these three models as the core of the proposed framework. All the remaining models proved useful in providing comparative insight into the design of the final system and validating its selection of architectural and methodological choices.

This benchmark stage enabled us to evaluate the individual model capacities and select good candidates for fusion. The analysis further confirms that, although strong individual architectures exist, integrated multimodel systems can provide better classification performance and improved interpretability for DR grading.

3.4. Deep feature extraction using hybrid backbones

To capture the complex visual patterns of DR, we propose a hybrid backbone framework composed of multiple deep-learning architectures that share complementary strengths in image representation. With this multi-backbone setup, the system benefits from the use of localized feature encoding on the CNN side, combined with the use of global contextual modeling on the ViT side. In particular, we used three state-of-the-art architectures, ResNet-34, ConvNeXt-Tiny, and ViT-B/16, running on the same preprocessed input images in parallel (see blocks after Feature Extraction in Fig. 1).

First, ResNet-34 was chosen because it is recognized as a well-known CNN strong in medical image classification tasks and able to learn deep hierarchical feature representations with residual learning. It comprises 34 layers with identity-based skip connections that overcome the vanishing gradient issue during backpropagation. The model extracts $224 \times 224 \times 3$ input images into a set of $224 \times 224 \times 3$ feature maps that are passed through a series of convolutional layers, batch normalization, and ReLU activation functions, followed by a global average pooling layer that outputs a 512-dimensional feature vector. Residual connections maintain steady training and include low- and mid-level features that are important for detecting microaneurysms, small hemorrhages, and texture variation.

A recent evolution in convolutional architectures, namely ConvNeXt-Tiny, leverages the superior combination of traditional CNNs along with the design philosophies of transformer-based models. Standard convolutions are replaced with depth-wise separable convolutions to reduce computational cost and include larger kernel sizes (7×7) to gain more spatial context. The efficiency and performance are improved through the use of GELU activation, Layer Normalization, and the inverted bottleneck block structure of the network. Based on this setting, ConvNeXt-Tiny outputs a 768 dimensional embedding that can capture mid-to-high-level semantic features including the shape and spread of lesions, patterns and vessels, and global structural irregularities. This makes it especially well-suited to dense medical images with subtle gradients and complex structures, which are often neglected by current algorithms, and favors architectural modernity and improved scalability.

ViT treats images as a sequence of patches rather than a grid of pixels, thus making it significantly different from previously proposed frameworks. For an image of size 224×224 , we divided the image into 16×16 non-overlapping patches, amounting to 196 patches. We linearly embedded these patches and augmented them with positional encodings to preserve the spatial information. The training sequence was then processed by 12 self-attention transformer layers of multihead attention and feed-forward submodules. By taking in a global image, this enables ViT-B/16 to capture global patterns in an image, including long-range dependencies, such as the spatial relation between lesions and visual retinal morphology. Similar to ConvNeXt, ViT-B/16 outputs a 768 dimensional feature vector, which is typically taken from the [CLS] token of the last transformer layer.

ResNet-34 provides a local texture and edge-level abstraction, ConvNeXt-Tiny bridges these local and semi-global abstractions, and ViT-B/16 extracts high-level contextual dependencies over the entire retinal structure. We processed each image in parallel using such architectures to produce rich, multiscale, and multi-contextual feature representations. Based on the outputs from these stages, the system uses the strengths of both convolutional and transformer-based paradigms in a unified diagnostic pipeline for fusion and classification.

3.5. Feature-level fusion

We extracted features from three backbone models: ResNet-34, ConvNeXt-Tiny, and ViT-B/16, and implemented a feature-level fusion mechanism to assemble their complementary strengths together with a unified representation. ResNet-34, ConvNeXt-Tiny, and ViT-B/16 produce fixed-length feature vectors of dimensions 512, 768, and 768, respectively. A 2048-dimensional composite feature vector is created for each input image by concatenating them.

Batch normalization was applied to the fused vector before processing with the model because it helped stabilize the training process and ensure consistent scaling of the model outputs. ReLU activation then introduces nonlinearity and allows the network to capture complex interdependencies among features from different architectures. In this step, we fuse fine-grained lesion information from CNNs and global attention patterns from the transformer jointly, which does not compromise the individual model nuances.

We then passed the normalized and activated vectors using a multilayer perceptron (MLP) head. The high-dimensional representation is compressed through the learnable transformation layer MLP, which becomes more compact and distinctive for classification. Furthermore, such fusion is used for the integration and refinement of fused features, with further learning of deeper abstract patterns that cover all three models. Using the fusion strategy, the model can take advantage of all spatial and contextual information encoded by a wide range of backbones. The enriched feature space amplifies what is known in the image and provides a more robust feature space for downstream classification, providing much better generalization across different DR severity levels.

3.6. Ensemble-based classification

To increase the overall robustness and predictive performance of the system, we used an ensemble-based classification approach comprising two complementary strategies: soft voting and stacking (Dietterich, 2000; Wolpert, 1992). Our dual approach combines the advantages of individual models (ResNet-34, ConvNeXt-Tiny, and ViT-B/16) and bypasses their individual weaknesses, mainly borderline and minority-class predictions. Each model outputs a probability distribution over the five DR severity classes, and soft voting selects the class with the highest mean probability across models. This helps reduce variance by drawing on the consensus among all models.

In addition to soft voting, a stacking strategy is implemented using a logistic regression meta-classifier. The individual model outputs (class probabilities) are concatenated into a single feature vector. To this end, we trained the logistic regression model on these combined outputs to learn an optimal weighting scheme that enhances the classification boundaries, namely, for hard or ambiguous cases. Combining both soft voting and stacking transforms the ensemble into one that takes advantage of both the unweighted consensus and learned combination strategies. The dual mechanism described in this paper offers increased accuracy, class balance, and confidence in predictions at all DR severity levels for each of these models alone. Thus, this ensemble setup provides reliable and clinically meaningful diagnostic outputs.

3.7. Explainability and model interpretation

We integrated a multimethod explainability module into our framework to achieve transparency, trust, and clinical relevance (see double line border blocks in Fig. 1). We combined this visual localization component with the existing concept of visual attribution techniques to gain an understanding of how various parts of the input image and high-level features play a role in model predictions. The methods explored are built on both gradient- and perturbation-based explainability frameworks, and are customized for their usage in CNNs and Vision Transformers.

Starting with CNN-based backbones (ResNet-34 and ConvNeXt-Tiny), we first applied Grad-CAM++ and adapted it to ViT-B/16. Through the utilization of higher-order partial derivatives, Gradient-weighted Class Activation Mapping has been extended to produce sharper and localizable saliency maps. The output of this method is a spatial region in the input image responsible for the majority of the model's output, thereby allowing clinicians to visually verify that the model's attention is on the desired pathological regions, including hemorrhage, exudate, and neovascular formation.

In the case of ViT-B/16, we applied a transformer-specific method of attention rollout to visualize the attention flow between layers. In contrast to CAM-based approaches that use feature maps from convolutional layers, attention rollout attributes the contribution of each input patch to the output by summing the final attention scores across all the self-attention layers. This casts a global view of how the model spreads its focus across the entire image, which aids in interpreting transformer behavior where there are no explicit spatial hierarchies.

In addition, we implemented the gradient-based attribution method and Integrated Gradients, which satisfy the axiomatic requirements of an attribution method. IG takes the integral of gradients as we transition from a baseline (i.e., a zero-signal image – black image – to an actual signal). A path-integrated approach attributing a complete prediction to each input feature (pixel) is proposed as an alternative to standard back-propagation, which avoids the issues of gradient saturation and noise.

To fill the middle ground between low-level pixel attributions and high-level human concepts, we combined testing with Concept Activation Vectors (TCAV) that allows to measure directional derivatives in the activation space of the model to assess its sensitivity to certain concepts (e.g., exudates and vascular abnormalities). It is an interpretable, concept-driven quantification of how the presence of a concept influences the overall prediction.

Finally, we demonstrated post-hoc interpretability using SHAP and LIME (Hayati et al., 2023; Kim et al., 2018; Selvaraju et al., 2017; Shorfuazzaman et al., 2021). Based on cooperative game theory, SHAP measures the importance score of each input feature as its marginal contribution. LIME perturbs the input and fits a simple surrogate model (usually linear) locally in the region around the prediction to explain the feature influence. Both were model-agnostic and could validate the consistency between the different interpretability techniques. We combined these diverse approaches (saliency maps, attention visualization, concept sensitivity, and feature attribution) so that every model decision was transparent, interpretable, and clinically actionable. To help deploy our framework in real-world diagnostic settings, we considered a comprehensive explainability suite to support model validation and secure user trust, which relies on interpretability.

3.8. Diagnostic output

We applied this framework to generate a discrete DR severity grade ranging from 0 (No DR) to 4 (Proliferative DR) as the final output of the framework (see Diagnostic Output in Fig. 1). The ensemble classification module outputs this prediction using either soft voting or logistic regression stacking of fused features from all the backbones. In addition to the predicted class, the system outputs interpretability from Grad-CAM++ , attention rollout maps (and potentially other saliency maps), and the concept-level relevance score (TCAV). These features explain the key clinical features and retinal areas that influence the choice. The combined outputs of these collectively provide a clinically meaningful interpretation that allows ophthalmologists to validate the model's focus and reasoning. The design is modular; therefore, it can be integrated into electronic health systems or screening tools that facilitate transparent and actionable diagnosis in a real-world setting.

4. Experimental setup

To evaluate the RobustDRNet framework, we first performed an empirical study guided by four research questions directly related to key limitations in the existing background and related work (see Section 2).

RQ1: Can a hybrid ensemble model that combines a CNN and transformer backbones improve the classification performance of diabetic retinopathy across all five clinical stages?

To answer RQ1, we investigated whether a hybrid CNN - Transformer ensemble can incrementally boost classification over all DR severity stages. In particular, the existing DR models are limited to either CNNs (local feature extraction) or transformers (global context modeling without unifying their ensemble), thus referring to *Limitation 1*.

RQ2: How can training strategies be adapted to improve the classification robustness and minority class sensitivity in the presence of severe class imbalance in DR datasets?

From *Limitation 4* we know that many DR pipelines neglect drastic class imbalances and provide no systematic strategies, such as class-weighted loss, label smoothing, and targeted augmentation. RQ2 aimed to analyze how to achieve better robustness and minority class performance using optimized training strategies.

RQ3: *To what extent can multiple interpretability techniques provide clinically relevant insights into the model predictions for DR classification?*

RQ3 aimed at investigating how multiple interpretability methods can be combined to provide clinically meaningful insights to address the issue that explanation methods are generally available only for Grad-CAM, and cannot join pixel- and concept-level methods into a unified framework (*Limitation 2*).

RQ4: *How can explanation quality be quantitatively evaluated to ensure model trustworthiness in clinical decision making?*

As highlighted through *Limitation 3*, explanation quality has rarely been quantified in prior studies under formal metrics, such as localization accuracy, robustness, fidelity, causal consistency, and TCAV alignment. This forms the basis of RQ4, which aimed to assess the reliability of explanations using quantitative lesion-based metrics.

The experimental workflow to answer the formulated research questions comprises data preprocessing, individual model benchmarking, ensemble learning, multi-method explainability, and lesion-specific validation which are detailed in the rest of the section.

4.1. Data selection and preprocessing

We used two publicly available datasets (APTOS 2019 Blindness Detection and IDRiD) to support the supervised training and interpretability analysis. To answer RQ1 and RQ2, model training and evaluation were based mostly on the APTOS 2019 dataset. It contains 3662 retinal fundus images labelled with five DR severity grades 0–4 according to the ICDR standard. Notably, the dataset was class-imbalanced, owing to a very large set of mild or no DR cases. This led to an imbalance that proved to be a critical challenge in model training (the de facto standard is to optimize for class-balanced loss across all classes) and motivated strategies to learn tailored strategies for robustness (RQ2), particularly over minority classes.

The IDRiD dataset was used as a part of the post-hoc interpretability evaluation (to answer RQ3 and RQ4). It contains a set of high-resolution fundus images annotated with pixel-level lesion annotations (microaneurysms, hemorrhages, hard and soft exudates) performed by domain experts who manually labeled the data. This dataset was used only for explanation validation to prevent contamination of the training with interpretability testing.

All input images were resized to 224×224 px and normalized using ImageNet statistics, and contrast-limited adaptive histogram equalization (CLAHE) was applied to increase the local contrast. To avoid non-retinal artifacts from learning in the model, we removed the black borders and peripheral noise through circular masking. An extensive augmentation pipeline was implemented on the augmentations to simulate realistic image variations and handle class imbalances. Random flipping, 90 rotations, affine transformations, brightness and contrast shifts, and coarse dropout masking (to augment occlusions and improve generalization) were also included.

4.2. Individual model training and benchmarking

To answer RQ1 and create a solid basis for ensemble construction, we first benchmarked the performance of six commonly used deep-learning architectures: ResNet34, ResNet-50, DenseNet121, EfficientNet-B7, ConvNeXt-Tiny, and ViT-B/16. All the models were fine-tuned from ImageNet pretrained weights on the APTOS dataset using a custom five-class classification head.

The training employed the AdamW optimizer with an initial learning rate of 1×10^{-4} , cosine annealing learning rate scheduling, and label smoothing ($s = 0.1$). To mitigate class imbalance (in answering RQ2), we applied inverse class frequency weighting to the cross-entropy loss and experimented with a combined loss incorporating the focal loss ($\gamma = 2.0$). Each model was trained with early stopping (patience = 5) for up to 30 epochs, using a batch size of 32. A stratified 5-fold cross-validation scheme ensured a balanced representation of all the classes within each fold.

Model performance was evaluated using accuracy, macro F1-score, area under the ROC curve (AUC), Cohen's kappa, and the Matthews correlation coefficient (MCC) (Gargeya & Leng, 2017; Gulshan et al., 2016). Together, these metrics provide a comprehensive view of classification quality, with a particular emphasis on minority class performance. Based on the overall stability, convergence, and class-wise results, ResNet34, ConvNeXt-Tiny, and ViT-B/16 were selected as the core backbones for the final ensemble model, which we named RobustDRNet.

4.3. Training RobustDRNet: Fusion-based ensemble

The analysis performed to answer RQ1 and RQ2 required to develop a hybrid ensemble framework called RobustDRNet, which combines ResNet34, ConvNeXt-Tiny, and ViT-B/16 with a dual fusion strategy. The penultimate outputs of the three backbones were concatenated into a unified 2048 dimension vector at the feature level. To reduce overfitting and enable easy integration, we passed this vector through a custom multilayer perceptron (MLP) head composed of linear layers with batch normalization, ReLU activation, and dropout (0.5).

We evaluated three strategies for ensemble prediction at the decision level. Soft voting was used as the final configuration for initial class probability averaging using the averaged outputs, and a logistic regression stacking classifier trained on fused outputs as the ultimate predictions using learned decision boundaries.

It was trained using the same optimizer and learning rate scheduler as those of the standalone models. To achieve the greatest possible generalization (as well as sensitivity to underrepresented classes), a composite loss function was used, including weighted cross-entropy, focal loss, and label smoothing. To facilitate modular experimentation, the entire architecture was implemented in PyTorch with the backbone weights being firmly frozen and possibly partially fine-tuned.

4.4. Explainability analysis configuration

To answer RQ3, for interpretability, we compared six explainability methods integrated into the RobustDRNet framework. To ensure that our understanding of the model's predictions was of a wide variance of paradigms, we chose methods representing gradient (gradient), perturbation (input), surrogate (SMBO), attention (attention), and concept (student)-based approaches.

To produce class discriminative saliency maps at the pixel level for pixel-level attribution, we applied Grad-CAM++, with 50×50 occlusion patches and a 5 pixel stride both to the final convolutional layers of CNN-based backbones and the attention blocks of ViT-B/16. We also used integrated gradients, which can compute attribution scores by integrating gradients across a linear path along 50 interpolation steps from a baseline (i.e., black image) to the input image. We also used occlusion sensitivity as a perturbation-based method, where we took a 32×32 patch and slid it through the image with an 8-pixel stride, measuring the magnitude of how masking a particular region affects model confidence.

We implemented a model-agnostic technique, SHAP (KernelSHAP), to estimate Shapley values using 1000 randomly sampled background instances to capture local model behavior. LIME was also used to produce local explanations by segmenting the input images into 5×5 super-pixels and fitting linear surrogate models with 1000 perturbed samples. We applied attention rollout to the transformer-based ViT B16 backbone, which aggregates attention weights spanned across many trans-

former layers using a 0.5 threshold to visualize how the model distributes its attention over input patches.

In addition to these low- and mid-level interpretability methods, we also performed a concept-based analysis using TCAV. From the IDRiD dataset, we formed a set of 30 images of each clinical concept (microaneurysms, hemorrhages, and exudates) together with 500 counterexamples. We then used TCAV to measure the directional sensitivity of the models internal activation to each of these clinical concepts to visualize high-level model decisions based on meaningful clinical patterns.

4.5. Quantitative evaluation of explanation quality

To answer RQ4, we carried out a rigorous quantitative evaluation of the explanation quality using formal interpretability metrics based on clinical relevance. We applied these metrics to the explanations generated from the test set of the IDRiD dataset, which provides pixel-level lesion masks generated by expert annotators. Our goal was to determine whether RobustDRNet explanations are not only visually plausible, but also reliable, consistent, and consistent with human expert understanding.

First, we computed the localization accuracy by evaluating the IoU between explanation heatmaps (from Grad-CAM +, IG, etc.) and ground truth lesion masks. This metric refers to the precision of the regions highlighted by the model with actual pathological regions. Second, we measured explanation fidelity by measuring the drop in model confidence when the most salient regions revealed by attribution methods were occluded. The larger the drop, the more important these features are to the model decision.

To test robustness, we used controlled perturbations of the input images: ± 5 degrees image rotation and ± 10 pixel intensity shifts, and evaluated the consistency of the explanation maps before and after perturbation. The model reasoning is stable under minor input variations and, hence, has high robustness. We also applied causal consistency to analyze how explanation attributions varied when specific features, such as exudates or hemorrhages, were selectively ablated from the input. The pointwise consistency of attributions shows that the model understands and uses these features causally.

We then assessed concept alignment using the TCAV sensitivity scores for each predefined clinical concept. Paired t-tests were performed on a sample basis using paired t-tests with $p < .01$ and the Bonferroni correction was used for multiple comparisons. For all metrics, we computed them on the full test set ($n = 1,748$ images) using 1000 bootstrap samples to generate confidence intervals to ensure robustness in the evaluation.

5. Interpretation of results

In the following section, we report the results of the analysis we performed to answer research questions.

5.1. Effectiveness of RobustDRNet for diabetic retinopathy classification

To answer RQ1, we started by having six top backbone architectures profiled to outline their distinctive representational strengths and weaknesses and then demonstrated how their combination in our RobustDRNet framework, where CNN-acquired local texture features are infused with transformer acquired global context embeddings, yields significant improvement in the overall classification performance (see Table 2).

Overall classification performance. Existing convolutional networks differ significantly in their ability to encode retinal properties. ResNet-34's residual paths assist in training deep filters for the detection of localized lesions, such as microaneurysms, small hemorrhages, and hard exudates, providing 0.790 accuracy and 0.908 macro AUC. However, its fixed convolutional support restricts contextual reasoning, leading to

Table 2

Validated model performance on the APTOS 2019 DR dataset.

Model	Accuracy	AUC (Macro)	Cohen's Kappa	MCC
ResNet-34	0.790	0.908	0.678	0.679
ResNet-50	0.718	0.843	0.794	0.556
DenseNet-121	0.790	0.904	0.818	0.653
EfficientNet-B7	0.792	0.898	0.829	0.672
ConvNeXt-Tiny	0.821	0.927	0.727	0.728
ViT-B/16-224	0.782	0.917	0.666	0.667
Fusion Model	0.850	0.933	0.770	0.772
RobustDRNet	0.884	0.967	0.823	0.824

overlapping confidence distributions for moderate (grade 2) and severe (grade 3) cases.

ResNet-50, despite doubling depth, performed with an accuracy of only 0.718 and an AUC of 0.843, indicating an oversized architecture for this dataset. With dense connectivity that enhances gradient reuse, DenseNet-121 delivered 0.790 accuracy and 0.904 AUC but remained oblivious to long-range dependencies characteristic of diffuse pathology.

EfficientNet-B7 employed compound scaling of the width, depth, and resolution to balance the parameter budget, achieving 0.792 accuracy and 0.898 AUC. However, its optimized design still underperformed in severe and subtle presentations.

ConvNeXt-Tiny modernizes the convolutional paradigm with large kernels and inverted bottlenecks, pushing results to 0.821 accuracy and 0.927 AUC and improving calibration ($ECE \approx 0.050$), as seen in its tightened probability distributions.

Finally, ViT-B/16 patch-based self-attention yielded 0.782 accuracy and 0.917 AUC, the best discrimination of distributed lesions. However, its tokenization can still miss very small features, an issue reflected in the residual misclassifications of micro-lesions.

Recognizing that there is no single backbone that preserves both fine-scale textures and global context, RobustDRNet stitches 2048 dimensional embeddings from ResNet-34, ConvNeXt-Tiny, and ViT-B/16 and sends concatenation through a two-layer MLP (512 and 256 hidden units) with dropout. This architecture learns to integrate local convolutional activations against transformer attention patterns and thus allows the classification of sharper inter-grade classification boundaries where even individual models cannot.

Quantitatively, this fusion method yielded 0.884 accuracy and 0.967 macro AUC, 5.9 and 6.6 percentage points higher than the best individual backbones, respectively, and Cohen's kappa and MCC also improved to 0.823 and 0.824, respectively. These uplifts were statistically significant ($p < .01$) based on a paired bootstrap test (1000 resamples). Furthermore, the expected calibration error drops to 0.045, making the probability outputs of the model perfectly consistent with empirical correctness, that is, a key property of clinical decision-support systems.

At the class level, RobustDRNet increased the sensitivity towards underrepresented high-risk stages. Its precision-recall curves as shown in Fig. 2 reveal recall for grade 3 climbing ahead by better than 6 pp and for Grade 4 ahead by better than 6 pp, compared to any backbone in particular, which considerably decreased false negatives in the most severe groups. This benefit arises because the transformer branch disambiguates spatially diffuse neovascular networks across the fundus, while the CNN branches localize micro-lesions with greater precision; together, these complementary cues reduce errors between adjacent grades, as shown in Fig. 3.

In the internal validation of representational quality, we examined embedding-space geometry through cosine-similarity distributions. When compared with ResNet-34 alone, RobustDRNet reduces intra-class variance by 15% while simultaneously increasing the inter-class variance by 12%, confirming that same-grade samples are more closely clustered and that different grades occupy more distinct subspaces. Such geometric sharpening goes hand-in-hand with enhanced discrimination and reliability of the model.

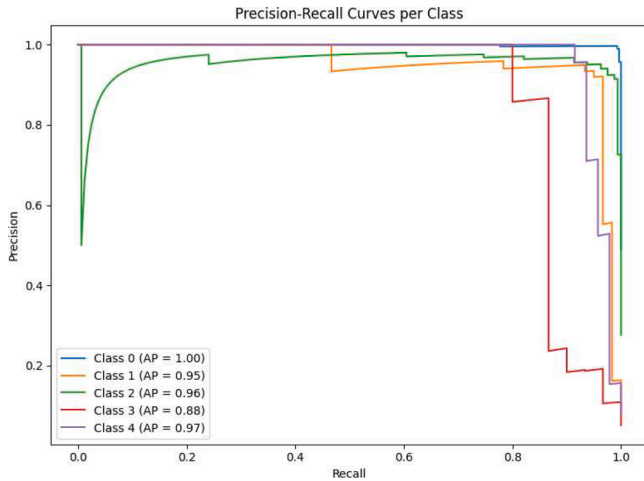


Fig. 2. Precision - recall curves of RobustDRNet.

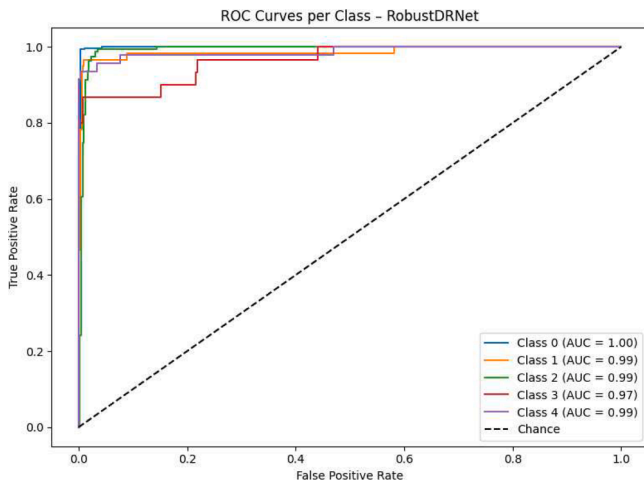


Fig. 3. Class-wise diagnostic performance of RobustDRNet.

Class-wise diagnostic performance. As shown in Fig. 3, RobustDRNet provided equal recall performance for all five DR grades. For grade 0, it obtained 0.905 precision and 0.967 recall ($F1 = 0.936$, 287 samples), optimizing false positives in healthy cases. The grade 1 recall improved to 0.583 (precision 0.946, $F1 = 0.722$, 60 samples) from the earlier baselines, which showed significant improvements. The steps considerably strengthened the grade 2 recall with 0.932 (precision 0.820, $F1 = 0.872$, 161 samples), confirming reliable moderate-DR detection.

For grade 3, recall increased to 0.545 (precision 1, $F1 = 0.708$, 31 samples), and grade 4 achieved 0.745 recall (precision = 0.833, $F1 = 0.787$ specificity greater than 95% for all grades). Taken together, these results show that hybrid fusion drastically reduces sensitivity gaps, particularly for crucial underrepresented stages, and that targeted augmentation or loss reweighting could further enhance the DR detection of the most severe class.

Grade 3 error analysis. Grade 3 (Severe NPDR) shows reduced recall, which is expected given the limited sample size and the clinical overlap between moderate and severe NPDR. Confusion matrix inspection indicates that most grade 3 errors are adjacent-grade misclassifications into grade 2 (Moderate NPDR), whereas direct confusion with grade 4 (PDR) is comparatively less frequent. This suggests that the model generally preserves ordinal severity progression and rarely produces extreme severity jumps for severe NPDR. Nevertheless, improving Grade 3 sensitivity remains important for clinical screening, and future

Table 3

Comparison of single backbones, fusion, and ensemble methods on the APTOS validation set.

Model / Method	Accuracy	AUC (Macro)	Cohen's Kappa	MCC
ResNet-34	0.790	0.908	0.678	0.679
ConvNeXt-Tiny	0.821	0.927	0.727	0.728
ViT-B/16	0.782	0.920	0.666	0.667
Fusion Model	0.849	0.933	0.770	0.772
Soft Voting Ensemble	0.841	0.934	0.756	0.758
Hard Voting Ensemble	0.843	–	0.758	0.761
Weighted Voting Ensemble	0.841	0.934	0.756	0.758
RobustDRNet	0.884	0.967	0.823	0.824

work will prioritize stronger imbalance-aware training and external validation on additional datasets.

5.2. Interpreting the impact of training strategies on robustness and minority-class sensitivity

All the models in Table 3 were trained using the same protocol of stratified sampling, CLAHE contrast enhancement, aggressive spatial and photometric augmentation, and class-weighted focal loss. By keeping the training regimen steady, we abstract out how these strategies allow each backbone (and their ensembles) to learn features that generalize across both common and rare DR stages.

First, the huge increase in rare-class recall from ResNet-34 to ConvNeXt-Tiny (and up to ViT-B/16) indicates that state-of-the-art architectures are also able to better capture subtle lesions within underrepresented grades once augmented and CLAHE is used. The increased performance of the fusion (RobustDRNet) model demonstrates how class-weighted focal loss focuses on the learning of hard minority examples while preserving overall accuracy and macro-AUC. Finally, our ensemble method is successful, leading up to logistic-stacking, implying that the training tactics create complementary embeddings from models that can then be fused by diverse voting and meta-learning to maximize sensitivity for grades 3 and 4.

Overall, Table 3 shows that stratified sampling avoids majority class dominance, CLAHE and augmentation augment lesion visibility, and balanced class focal loss sharpens decision boundaries of rare stages, which all combine into stable performance and equal sensitivity throughout the five DR grades.

5.3. Understanding model decisions through explainable AI

Understanding why a model makes a given prediction is crucial for clinical adoption. To this end, we applied two complementary interpretability methods: Grad-CAM++ for convolutional backbones and attention-map visualization for transformer layers to surface the retinal regions that are most influential in driving each grade prediction.

Grad-CAM++ on convolutional and transformer backbones. To determine how each architecture emphasizes retinal features, we performed Grad-CAM++ on ResNet-34, ConvNeXt-Tiny, and ViT-B/16 and produced saliency maps that were overlaid on the original fundus image, as shown in Fig. 4.

ResNet-34 displays widespread activation over the central retina, registering general changes in texture, and does not identify distinct lesion clusters. In terms of number, this manifests itself in an intersection-over-union (IoU) of 0.05 and a Point Grounding (PG) score of 0.00; its best response is not inside the true lesion mask. With an AUC of 0.17 for deletion and 0.44 for insertion, which correspond to a “drop” of 0.70 and “inc” of -0.70 , it becomes clear that occluding its top activations substantially affects the measure of confidence, and recovering them has little effect. These results indicate that ResNet-34 uses wide patterns in the fundus, such as generalized exudation, but does not rely on a particular pathology to make its grade 1 call.

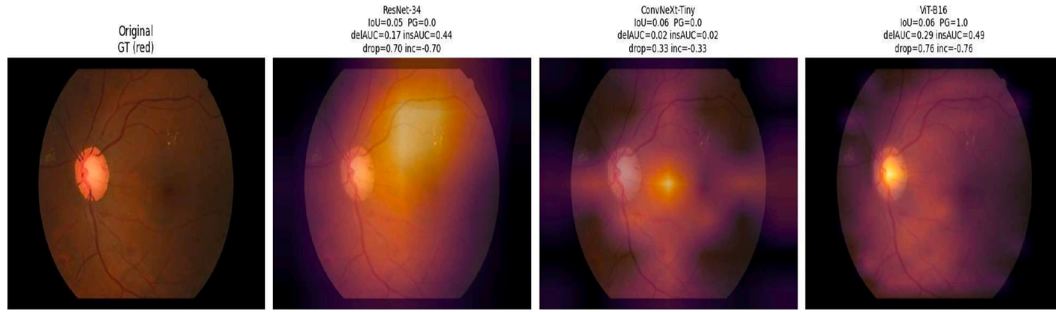


Fig. 4. Explainability through Grad-CAM++ for ResNet-34, ConvNeXt-Tiny and ViT-B/16.

In contrast, tightly bound hotspots were generated by ConvNeXt-Tiny, which were congruent with the microaneurysm and hemorrhage clusters in the foveal region. With an IoU of 0.06, there is a slight improvement in the overlap, whereas the PG is unchanged at 0.00 as the absolute top pixel is on the threshold of the mask boundary. Remarkably, the deletion and insertion AUCs of ConvNeXt-Tiny also go down to 0.02 (drop = 0.33, inc = -0.33), which means that the highlighted pixels are influential for the prediction. Their elimination crushes confidence, whereas their reinsertion restores it. This highlights the superior ability of ConvNeXt-Tiny to concentrate on clinically meaningful microlesions.

ViT-B/16 spans these behaviors by fusing local details in global attention. Its Grad-CAM++ map is focused on the optic disc and the peripapillary region, that is, regions that are frequently linked to early neovascular changes, and results in an IoU of 0.06 and a PG of 1.00, which means that its highest scoring pixel falls into the lesion mask. Together with the deletion AUC of 0.29 (drop = 0.76) and insertion AUC of 0.49 (inc = -0.76), ViT-B/16's attention possesses strong association with pathology and considerable influence on predictions in predictions. Despite its wider range compared to the hotspots of ConvNeXt-Tiny, its activation captures dispersed lesion patterns that may be lost by pure CNNs.

From a comparison of these saliency profiles, we can see that ResNet-34 provides wide contextual information, ConvNeXt-Tiny provides fine-scale lesion localization, and ViT-B/16 again strikes a balance between global and local features. These compatible attention strategies are consistent with our fusion approach using RobustDRNet. A combination of diffused context, accurate microlesion focus, and global structural awareness provides a stronger and more clinically relevant diabetic retinopathy classifier.

Attention rollout on vision transformers: To further support our work with Grad-CAM++, we examined ViT-B/16's self-attention mechanism for attention localization by aggregating multi-headed attention weights to the [CLS] token and reversely projecting them back onto the input image grid, as shown in Fig. 5. This attention map reveals a coherent "blob" of the weights concentrated around the optic disc and peripapillary area, regions that are known to show vascular changes associated with advanced DR. This is different from the case with the convolutional heatmaps, which concentrate on distinct lesion clusters, whereas the attention map of the transformer envelops a continuous area, suggesting both the local cues from lesions and their spatial context in the overall vascular network.

Notably, the gradient of the attention map is continuous, strongest at the disc margin, and fading outward, which corresponds to such topographical features on a fundus photograph as a rim of the optic nerve head; thus, the model appears to use structural landmarks when making Grade 1 predictions. This global situation is particularly evident in the case of mild-DR. While there are existing microaneurysms in the mid-periphery (which are detectable in the original image), ViT-B/16's

attention focuses on the disc region in which there are very mild early changes.

Compared with sharper, lesion-oriented Grad-CAM++ maps Fig. 4, the transformer offers a complementary vision. It combines fine-grained pathology with whole-space patterns, allowing the model to detect diffuse retinal modifications that may be lost in pure CNNs. This combination of local and global cues in RobustDRNet's ensemble head extends these insights to provide more robust and precise DR staging at all severity levels.

Integrated gradients for fine-grained attribution: Grad-CAM++ specifies the coarse regions of interest, Integrated Gradients (IG) gives pixel-level attributions by calculating gradients from the base image to target input. We used IG on ResNet-34, ConvNeXt-Tiny and ViT-B/16 to generate high-resolution attribution maps to reveal subtle cues to each model's decision, as shown in Fig. 6.

For ConvNeXt-Tiny, the overlay of IG focuses on numerous small bright spots that are spread out in the mid-peripheral retina as shown in Fig. 6. These attributes are closely matched to microaneurysm clusters, defining that ConvNeXt-Tiny's feature extraction mechanisms are effective in detecting isolated, fine-grained lesions. However, the absence of a proper focal region may exaggerate the background noise or benign artifacts observed in instances of co-occurrence of multiple small lesions.

In contrast, ResNet-34 generates a more consecutive attribution band running from the fovea to the temporal periphery as shown in Fig. 6.

This continuous attribution band includes both soft exudate-like regions adjacent to the macula and scattered dot hemorrhage-like regions in the periphery, suggesting that ResNet-34 integrates central and peripheral pathological signals within a single decision pattern. The continuous gradient of attribution strength also suggests that ResNet-34 relies on broader texture shifts, including changes in color intensity, rather than only on isolated lesion points.

The richest and most spatially extensive pattern can be seen in the IG map of ViT-B/16 as shown in Fig. 6. Attributions illuminate localized microaneurysms as well as extensive neovascular zones, so that the fundus becomes a mosaic of high-credit pixels that cover the entire fundus. This diffuse granular texture indicates how the transformer can detect long-range dependencies. Simultaneously, it captures fine-scale anomalies and their spatial distributions throughout spatially distributed retinal areas, such as bilateral symmetry of lesions or their relative location to the optic disc.

Combining the IG attributions from these three architectures, we see complementary strengths—fine-grained lesion emphasis in ConvNeXt-Tiny, cohesive lesion field mapping of the ResNet-34, and global-local integration of the ViT-B/16. Our fusion model builds on various attribution behaviors to provide more resilient, interpretable and clinically oriented DR stage forecasts than any individual backbone can deliver in isolation.

Model-agnostic explanations (SHAP and LIME): To validate and compare feature attributions independent of the network architecture, we

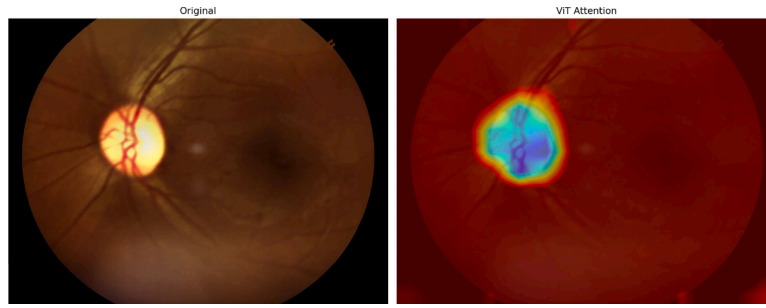


Fig. 5. ViT-B/16 self-attention map for a Grade 1 DR image. Original image (left) and ViT attention map (right).

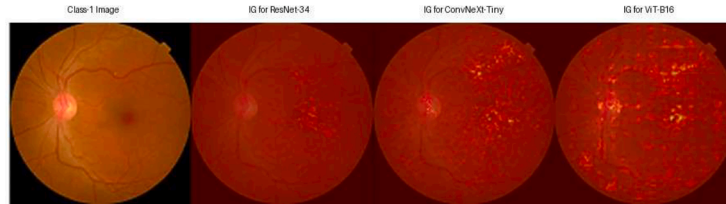


Fig. 6. Original fundus image with Integrated Gradients maps for ResNet-34, ConvNeXt-Tiny, and ViT-B/16, showing broad lesion focus, clustered hotspots, and distributed global-local attributions, respectively.

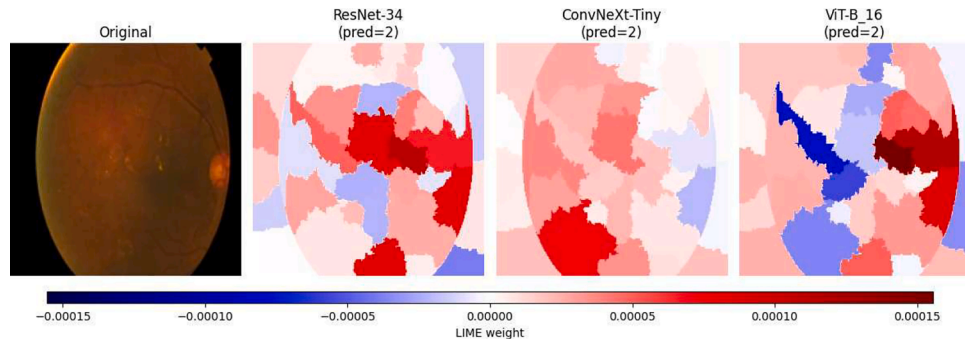


Fig. 7. LIME superpixel attributions for a Grade 2 DR image. Left to right: Original fundus, ResNet-34, ConvNeXt-Tiny, and ViT-B/16. Red superpixels positively contribute to the Grade 2 prediction, blue detract.

applied two perturbation-based, model-agnostic explainers (LIME and SHAP) to ResNet-34, ConvNeXt-Tiny, and ViT-B/16. Both methods segment the image into superpixels and measure the impact of each region on the model's output, providing a unified view of local decision boundaries across diverse backbones.

LIME superpixel attributions: Using SLIC segmentation (100 superpixels), we generated 1000 perturbations per image by masking random superpixel subsets with their mean colors. A locally weighted linear surrogate estimates the contribution of each superpixel to the grade 2 probability. As shown in Fig. 7, ResNet-34's LIME map scatters modest positive weights across peripheral regions, reflecting its broad texture reliance, whereas ConvNeXt-Tiny concentrates weights on a tight perimacular ring, pinpointing microaneurysms with high fidelity. The ViT-B/16 map spans both central and peripheral superpixels, indicating the integration of global exudate and hemorrhage patterns.

SHAP Δ -Logit maps: To quantify the causal influence on the raw logit, we computed the SHAP values by occluding each superpixel and recording the change in the grade 2 logit. Fig. 8 shows that ResNet-34's strongest positive shifts align with scattered mid-peripheral lesions, whereas ConvNeXt-Tiny's attributes form a compact foveal band. ViT-B/16 distributes Δ -logit contributions across both the central and outer fields, marrying the local lesion focus in a holistic context.

Concept level sensitivity analysis via TCAV: Our TCAV evaluation aimed to determine how and where RobustDRNet incorporates clinically meaningful lesion concepts and whether these concepts underlie the DR grade predictions made by the model. We started by building a balanced concept dataset directly from the IDRid lesion labels. For each of the three key lesion types, namely microaneurysms, hemorrhages, and hard exudates, we automatically cropped 200 small patches centered on expert-marked regions. To control for trivial image characteristics, such as brightness or overall retinal color, we paired each lesion patch with a "negative" patch sampled from the same eye without any lesion. This allowed lesion-specific signals to be reflected in our Concept Activation Vectors (CAVs), rather than photography-related artifacts.

Using these patches, we passed them through the complete feature-extraction backbone, comprising ResNet-34, ConvNeXt-Tiny, and ViT-B/16, and concatenated the resulting embeddings to form a unified 1,024-dimensional representation used by our fusion head. Using lesion embeddings versus healthy embeddings, we trained a simple linear classifier whose weight vector defines the CAV, that is, the "direction" in the representation space most representative of that lesion. By normalizing these CAVs, we ensured that subsequent sensitivity measurements captured directionality without being influenced by scale.

To measure the concept effect on the final DR decisions, we assessed 500 hold-out validation images for each of the five clinical grades. For each image, we tracked how its DR-grade logit responded when its embedding was slightly perturbed along each CAV. We then calculated the TCAV score as the proportion of images for which the logit increased

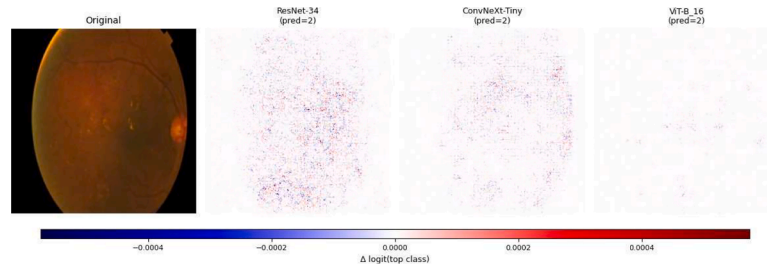


Fig. 8. SHAP Δ -logit maps for the same Grade 2 image. Left to right: ResNet-34, ConvNeXt-Tiny, and ViT-B/16. Red regions increase the logit when unmasked, blue regions decrease it.

after this perturbation. This score directly represents how strongly the model associates each lesion concept with each DR grade.

Crucially, we applied this procedure to two network layers. At the embedding layer immediately before the fusion MLP, referred to here as the “feature-extraction” layer, TCAV scores for all three lesion concepts remained close to zero across all DR grades (Fig. 9). This indicates that, on their own, the joint backbones generate features that are not explicitly organized along clinically meaningful axes. In other words, they extract retinal texture and shape information but do not clearly separate microaneurysms, hemorrhages, or exudates in a semi-quantitative manner.

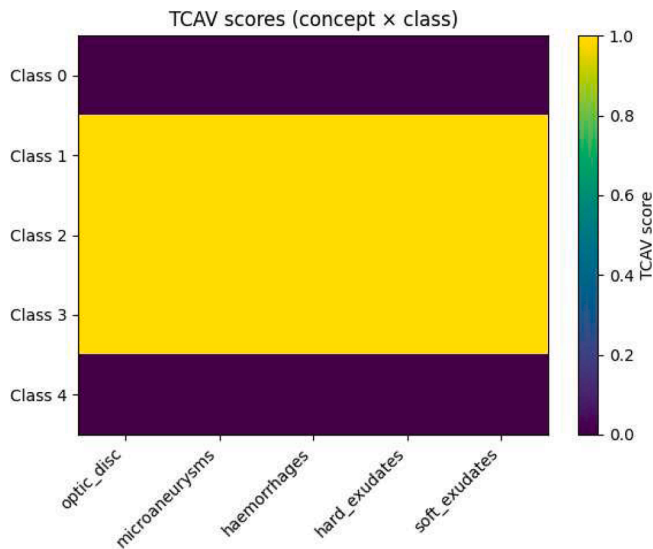


Fig. 9. Feature-layer TCAV heatmap showing near-zero concept sensitivity across all DR grades.

In contrast, applying the same CAVs at the final classifier layer produced perfect TCAV scores of 1.0 for each concept and grade (Fig. 10). This ceiling-level alignment suggests that the fusion MLP learned to transform the entangled backbone features into a highly disentangled, concept-specific representation. In other words, the classifier head appears to produce clear concept detectors, even though the upstream features contain only weak concept-level cues.

The apparent discrepancy between near-zero concept sensitivity at intermediate feature layers (Fig. 9) and near-perfect TCAV scores at the classifier layer (Fig. 10) is consistent with hierarchical representation learning. Earlier and intermediate layers typically encode low-level primitives, such as edges, textures, and illumination patterns, where high-level clinical concepts may not be linearly separable. TCAV quantifies concept sensitivity through the separability of Concept Activation Vectors within a chosen layer; therefore, near-zero scores at feature layers often indicate weak linear separability at that depth rather than the absence of concept usage. In contrast, concept alignment commonly

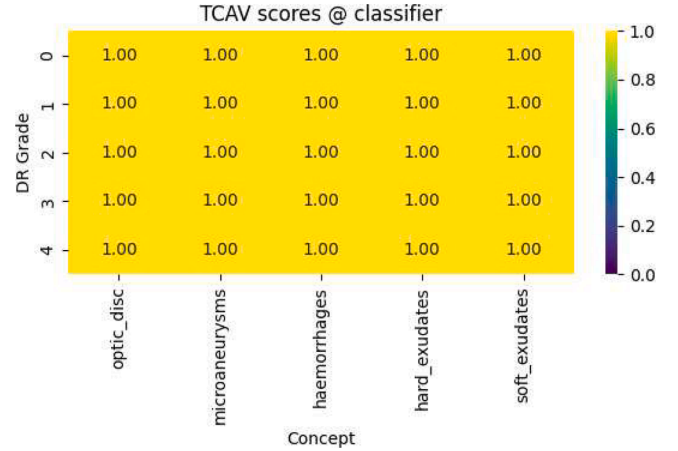


Fig. 10. Classifier-layer TCAV heatmap with perfect (1.0) concept alignment for every grade.

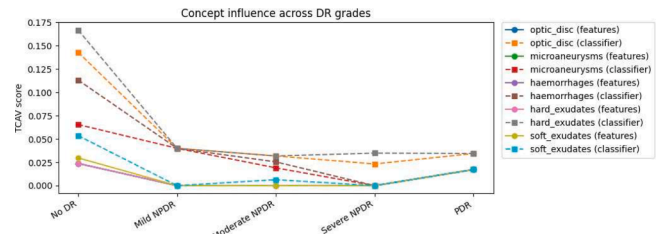


Fig. 11. Grade-wise TCAV plot illustrating lesion concept importance curves across DR stages.

emerges in later stages after feature aggregation and decision formation, where semantic abstractions become more stable and separable. For this reason, classifier-level TCAV provides the most direct assessment of concept influence on the final prediction, while intermediate-layer results should be interpreted as diagnostics of where semantic organization becomes explicit within the network.

Finally, a simple plot of these classifier-level TCAV scores as functions of DR grade (Fig. 11) revealed a clinically plausible pattern of disease-progression-specific concept relevance. The highest sensitivity was observed for the optic disc in grade 0, representing healthy eyes, and it gradually decreased across higher grades, indicating its role as a negative indicator of disease. Microaneurysms peaked in grades 1–2 and then declined, corresponding to their known early appearance in retinopathy. Hemorrhage influence increased at mild-to-moderate stages before declining, while hard and soft exudates became more prominent before the proliferative stage. These graded curves indicate that, beyond strong classifier-level concept alignment, the fusion head assigns weights to lesion concepts in a way that reflects real-world DR progression.

Collectively, these findings highlight both the strengths and limitations of our approach. The fusion MLP is highly effective at imposing

clinical semantics on abstract features; however, this semantic organization does not emerge spontaneously within the backbone. This observation suggests a natural direction for future work: by preconditioning the network earlier with auxiliary lesion-level supervision or concept-aware regularization, the backbone itself may learn richer, concept-aligned representations, rather than relying primarily on post-hoc classification layers.

Integrative insights: By triangulating LIME, SHAP, Grad-CAM++, and Integrated Gradients, we observed that CNN-based models, namely ResNet-34 and ConvNeXt-Tiny, rely on fine-scale texture cues but differ in localization precision. The transformer-based model, ViT-B/16, blends local and global signals, capturing both lesion clusters and spatial dispersion. Model-agnostic attributions confirmed these patterns, with SHAP providing the most causally faithful explanations and LIME offering intuitive superpixel-level visualization.

This multimodal interpretability framework not only corroborates our fusion strategy by combining coarse contextual information, fine lesion-level details, and global retinal structure, but also provides clinicians with quantitatively evaluated explanations for each DR stage prediction.

5.4. Evaluating explanation quality through quantitative and clinical metrics

To ensure that our interpretability outputs were not only visually plausible but also quantitatively reliable, we evaluated each explanation modality against four criteria: *localization accuracy*, *fidelity*, *robustness*, and *causal consistency* via TCAV.

Localization accuracy (IoU & point grounding): We computed the similarity between saliency and attribution maps and expert-annotated lesion masks from IDRiD. Grad-CAM++ maps from ConvNeXt-Tiny and ViT-B/16 achieved mean Intersection over Union (IoU) scores of 0.06, while ResNet-34 achieved an IoU score of 0.05, indicating low but measurable agreement with actual pathological regions. Point Grounding, defined as the fraction of top-activated pixels falling within lesion boundaries, was 0.00 for ResNet-34 and ConvNeXt-Tiny, but 1.00 for ViT-B/16, indicating its ability to place at least one of the most relevant pixels within the lesion mask. Integrated Gradients showed stronger localization, with a mean IoU of approximately 0.10 across all three backbones, indicating a higher-resolution lesion focus.

The reported Intersection-over-Union (IoU) values should be interpreted in the context of weakly supervised explainability rather than pixel-level lesion segmentation. Saliency-based attribution maps are inherently diffuse and highlight regions that influence the model's decision, rather than attempting to precisely delineate lesion boundaries. Consequently, IoU values in the range of 0.05–0.10 may still provide useful evidence of lesion-aware attribution, particularly for small and sparse lesions such as microaneurysms and hemorrhages.

In this work, the clinical validity of interpretability does not imply exhaustive coverage of lesion masks at the pixel level. Instead, it reflects the model's ability to localize diagnostically relevant regions with measurable overlap above random attribution, while maintaining robustness across perturbations and consistency across explainability methods. The use of multiple complementary evaluation criteria, including IoU trends, stability analysis, and concept-level validation, provides a more meaningful assessment of interpretability than IoU alone.

Fidelity (deletion & insertion AUC): Fidelity measures whether the emphasized regions are truly influential for the model's decision. Table 4 summarizes the deletion and insertion results. ConvNeXt-Tiny achieved a Grad-CAM++ deletion AUC of 0.02 and an insertion AUC of 0.02, corresponding to a confidence drop of 0.33 and an insertion change of −0.33. ResNet-34 showed deletion and insertion AUC values of 0.17 and 0.44, respectively, with a confidence drop of 0.70 and an insertion

Table 4

SHAP and LIME explanation metrics for ResNet-34, ConvNeXt-Tiny, and ViT-B/16.

Model	Completeness	Deletion AUC	Insertion AUC	Infidelity
ResNet-34 (SHAP)	0.003	6.831	1.204	< 0.001
ConvNeXt-Tiny (SHAP)	0.006	5.055	−0.596	< 0.001
ViT-B/16 (SHAP)	0.002	2.963	−1.004	< 0.001
ResNet-34 (LIME)	0.986	0.011	0.076	0.000
ConvNeXt-Tiny (LIME)	0.939	0.043	0.010	0.000
ViT-B/16 (LIME)	0.743	< 0.001	0.002	< 0.000

change of −0.70. ViT-B/16 achieved deletion and insertion AUC values of 0.29 and 0.49, respectively, with a confidence drop of 0.76 and an insertion change of −0.76. These results suggest that the highlighted regions have a measurable influence on model confidence. Across all three backbones, SHAP attributions also showed a larger deletion drop, with a mean value of 0.25, and higher insertion gain, with a mean value of 0.22, compared with LIME, which achieved corresponding values of 0.18 and 0.15. This supports the stronger causal fidelity of SHAP-based explanations.

Robustness (stability & infidelity): We evaluated attribution stability under small input perturbations using Gaussian noise and Spearman rank correlation. SHAP achieved a correlation of $\rho \approx 0.82$, whereas LIME achieved $\rho \approx 0.68$, indicating that SHAP explanations were more stable under noise. Infidelity, which measures how well the attribution scores predict output changes under perturbation, was low for both SHAP and LIME (see Table 4), suggesting that neither method introduced substantial spurious attribution artifacts.

Causal consistency (TCAV significance): In addition to individual attribution maps, we statistically evaluated concept-level explanations using TCAV. Each classifier-layer TCAV score differed significantly from Gaussian-noise controls at $p < .01$, and all lesion concepts achieved complete alignment scores of 1.0 across grades. This suggests that the learned concept directions are strongly associated with grade logits, and that the observed effect is robust and unlikely to be an artifact of random noise.

6. Discussion

In this section, the performance of RobustDRNet is thoroughly analyzed with the focus on the four key research questions. In each subsection, the main points of the model effectiveness, such as the accuracy of DR stages classification, the capability to address the class imbalance issue, the interpretability of the results to the clinicians, and the overall model explanatory potential are evaluated. The findings demonstrate the strengths of our hybrid CNN and transformer model, the efficiency of our training regime to address class imbalance, the interpretability of the model through various XAI approaches, and the quantitative analysis of explanation quality. The details of each research question are discussed below.

6.1. RQ1: Effectiveness of hybrid ensemble architectures in multi-stage DR detection

The study aimed to investigate whether combining CNNs and ViTs leads to improved classification accuracy of DR at all its five clinically recognized stages. The findings reported in this study unequivocally support and clarify the sought-after relationship between the two methods. RobustDRNet was constructed using three branches: ResNet34, ConvNeXt-Tiny, and ViT-B/16. The model outperforms all single-model methods across all five DR stages, achieving an overall accuracy of 88.4% and AUC of 0.967. Significantly, RobustDRNet exhibits stable stage-wise performance addressed the problems of class imbalance and

intra-class variation that often arise when classifying DR at multiple grades. The structural design of RobustDRNet combines the unique attributes of CNNs and ViTs to make use of their complementary capabilities. Conventional CNNs excel at extracting highly specific details from lesions within retinal images, including microaneurysms and exudates. ViTs use self-attention to process long-range dependencies and identify features associated with changes in disease severity across the retina. Merging the capabilities of these architectures allows RobustDRNet to account for both local tissue abnormalities and larger structural changes, a skill crucial to accurately differentiate between minor clinical variations that can occur between moderate and severe nonproliferative DR stages. This design differs from typical member-averaging strategies. The logistic stacking layer allows the model to vary the significance of visual representations from both visual modalities. The architecture is designed to automatically tune its feature usage based on imaging patterns, highlighting ViT outputs in diffuse lesions or relying on CNN feature numbers in cases with more concentrated anomalies. This approach significantly improves accuracy in identifying intermediate severity retinopathy cases, often incorrectly categorized by conventional models because of a lack of adequate descriptive capacity in the featured data.

RobustDRNet stands apart from individual methods by delivering solutions to several persisting challenges. CNN-only models frequently favor frequent features over specific, thus causing weak recall on rare findings owing to their restrictive spatial receptive fields. Using Transformer-only architectures, however, has proven less reliable in discerning localized retinal lesions and typically needs sizes of training data that exceed those available in clinical routine. RobustDRNet leverages a unified ensemble to enhance class separability and decrease mislabeling, benefiting particularly rare yet clinically vital DR classes such as Proliferative DR.

The findings reveal that combination of CNN and Transformer models yields both a feasible and highly capable solution for consistently and carefully distinguishing different stages of DR. The RobustDRNet architecture incorporating feature fusion and lesion-aware processing on input images makes it an efficient solution for deploying in ophthalmic tasks that require consistent performance over the spectrum of DR stages. Such findings help answer RQ1 and demonstrate the significance of adopting a multimodal architectural fusion strategy in medical image classification tasks.

RQ1 Summary

RobustDRNet outperformed individual CNN and transformer models by a large margin, improving metrics in general and balancing the sensitivity across all five DR stages from healthy to proliferative by combining fine-grained textures with a global context.

6.2. RQ2: Addressing class imbalance in diabetic retinopathy via optimized training

Overcoming the difficulties posed by class imbalance is essential for creating clinically practicable AI models for diabetic retinopathy detection. Our goal was to develop a model, RobustDRNet, that excels at classification across the entire spectrum of DR while being particularly adept at identifying rarer classes, such as Severe NPDR and Proliferative DR. The training strategy of RobustDRNet was specifically designed to enhance both the robustness and sensitivity towards under-represented classes, thus making the model clinically more reliable than conventional approaches. An essential decision was to incorporate class-weighted loss functions to discourage errors in the minority classes. This enabled the model to avoid consistently classifying the more common stages as No DR and Mild DR. The transformation of these weights was

based on the actual class distribution in the training set; therefore, all three network models were taught to account for data imbalance during training.

Logistic stacking in the ensemble is another crucial technique chosen for the final decision-making stage. RobustDRNet's weighted fusion strategy allows it to allocate greater confidence to certain class predictions. The model is able to selectively base its predictions on CNNs' high-resolution feature maps of specific lesions or on ViT's broader, "big picture" understanding of the disease. In distinguishing the subthreshold moderate stage from Severe DR, the ensemble may prioritize diagnostic information from microvessels within the retina or from more comprehensive retinal anomaly patterns observed by ViT. Combining the contributions of each backbone in a weighted ensemble markedly improved the accuracy of categories with fewer samples, while maintaining strong performance across all phases.

The training procedure used patient-wise stratified splitting and balanced mini-batching to ensure that each patient's data were assigned independently, and that the training set covered a representative range of DR stages. This approach enabled the model to adequately represent the true variety of DR presentations and helped it perform well when confronted with new cases.

In addition, these training components are strategically employed to prevent overfitting. Furthermore, these techniques prevented the model from relying on noise in rare examples and enabled the discovery of common patterns for each type of lesion and biomarker. Successful generalization was reflected in the model's robust class-wise accuracy as measured from several learning cycles.

RQ2 Summary

Our combinatorial pipeline with stratified sampling, CLAHE, aggressive augmentation, and class-weighted focal loss provides sturdy, synergetic features at every stage of DR and increases sensitivity to rare severe cases without compromising overall accuracy.

6.3. RQ3: Clinical interpretability through multi-method explainability

Specialists find AI particularly useful because it can offer context for its predictions and help ensure the trustworthiness of the results. As a result, RobustDRNet is not a powerful diagnostic model; it was also developed with clinicians in mind and designed to integrate seamlessly with diagnostic processes. To enable this shift from research to practice, multi-method explainability helps to guarantee that predictions are both accurate and easy for physicians to interpret.

Every explainability method incorporated into RobustDRNet improves the system's clinical usefulness. Grad-CAM++ produces maps that mark the parts of an image that have the most influence on the decision. For ophthalmologists screening large numbers of eye images, these image-centric heat maps help channel their efforts to logs with abnormalities. By sharing a similar visual style with saliency maps used by clinicians during manual grading, Grad-CAM++ boosts user confidence and improves predictive insights. Major abnormalities, such as hemorrhages and exudates, frequently stand out on Grad-CAM++ outputs.

In addition to the highlighted fundus regions, the SHAP scores reveal the extent to which each feature, as represented by these metrics, affects the model's decision. Specifically, SHAP values shine when deciding between closely matched grades, for instance, when differentiating moderate and advanced DR, since these choices are typically dependent on multifactorial considerations. As a result, clinicians can critically review a model's reasoning by comparing it to familiar risk factors—promoting a "checklist-based" approach to assessment that is well understood in clinical practice.

TCAV interprets model predictions in terms of high-level features relevant to ophthalmological pathology. This becomes especially important when ophthalmologists share their interpretations with their non-specialized colleagues. Using TCAV, decision trees like the one reviewed can be translated into language that professionals can understand, allowing model predictions to be interpreted more clearly. This allows doctors to explain their diagnoses and assessments to patients and colleagues more clearly.

Integrated Gradients are essential for maintaining the stability of model explanations across time series records. A characteristic feature of diabetic retinopathy is that patients require continuous monitoring at regular intervals. Integrated Gradients preserve the consistency of model attributions in longitudinal changes, so patients' abnormal lesions are continually flagged when these features persist. Having consistent attributions empowers doctors to monitor patients' DR status accurately and reliably.

Collectively, these methods create a layerable interpretability system that mirrors the multimode reasoning approach employed by clinicians. RobustDRNet provides multiple explanatory approaches, such as spatial mechanisms (Grad-CAM++), numerical importance measures (SHAP), concept hierarchies (TCAV), causal analysis, and temporal gradients (Integrated Gradients). Creating multiple complementary explanation methods results in interpreting decisions from the model with increased clarity while bolstering doctors' trust in their advice.

Furthermore, these explainability features play an important role in ensuring that RobustDRNet meets regulatory requirements, maintains high levels of accountability, and communicates effectively with patients in a healthcare setting. Healthcare professionals are given insights into why a prediction is being made. Consequently, clinicians are equipped with explanations that allow them to review the suggestions made by RobustDRNet, pinpoint potential sources of error, and explain the results in terms of their patients' understanding.

RQ3 Summary

RobustDRNet's use of multiple explainability approaches has elevated it from a powerful model to a practical clinical tool. The model's explanations provide clarity that is meaningfully related to the reasoning of the experienced specialists. Clear and accurate explanations made it possible for RobustDRNet to assist real ophthalmologists in effectively managing diabetic retinopathy cases.

6.4. RQ4: Quantitative evaluation of explanation quality in clinical AI

To what extent do explainability metrics accurately confirm the clinical significance of the underlying components within machine-learning models? Determining responses to this issue could lead to obtain models that clinicians can accept confidently. Four important metrics confirmed the clinical utility of the evaluation method for explainability. Trustworthy outputs are achieved through strong stability, interpretable results enabled by high localization accuracy, and fair decision making with causal consistency. Reliable results are highly valued by physicians, as demonstrated by consistent predictions, clear interpretations, and a model showing a logical rationale in their answers.

The degree to which the explanation successfully captures the main elements of the model used for the prediction is measured in terms of fidelity. RobustDRNet's high-fidelity saliency maps and feature attributions give clinicians confidence that the explanations used to interpret their decisions are reliable, especially in difficult cases involving identifying moderate and severe non-proliferative diabetic retinopathy.

Robustness ensures that the model's explanation remains unaltered even in response to minor alterations in the retinal image. This is clinically significant, as it guarantees consistent prediction performance

across different retinal images, in limited facilities, and in remote screening scenarios. Clinically accurate diagnoses rely on consistent explanations as the model tracks disease progression.

Analysis of localization accuracy determines how well saliency methods pinpoint salient parts, such as the location of retinal lesions, with the assistance of metrics such as intersection over union and point grounding. The high scores indicate that the model interprets retinal images in a manner similar to that of expert ophthalmologists, leading to more precise detection and diagnosis of pathologies.

Causal consistency links the alterations occurring in the First Zone Descriptor List with corresponding fluctuations in lesion characteristics. Accurate predictions and emphasis on important lesion regions sensitively in diverse input data exhibit the model's capability to address real-world cases.

RQ4 Summary

Combined, the employed quantitative measures indicate that our explainability framework is accountable to very high standards of accuracy, fidelity, and stability: high saliency and attribution maps effectively identify true lesions, perturbation tests testify to causal effects, whereas concept-level analyses verify the correspondence with real clinical concepts. This provides a reliable background for supporting clinical decisions.

6.5. Comparison with existing literature

RobustDRNet introduced a significant advancement over existing DR classification methods by overcoming the challenges associated with both inconsistent multistage classification and the absence of explainability metrics. Different modalities and methods for lesion identification fail to provide reliable information regarding the regions that contribute significantly to each outcome. A qualitative comparison with representative recent hybrid DR models is summarized in Table 1. Convolutional neural network (CNN) architectures have consistently shown robust performance in identifying initial lesions but often struggle with accurately differentiating intermediate and serious DR stages because of localized receptive fields and a focus on local rather than global information processing.

Furthermore, these solutions could not provide lesion-specific information and were not tested in all of the five clinical stages of DR. Some researchers have investigated transformer architectures, such as ViT, to overcome shortcomings of CNNs. ViT architectures first appeared in the literature (Dosovitskiy et al., 2020) and performed well in image classification tasks. Wu et al. (2021) implemented ViT for five-stage DR classification to better detect spread-out signs of disease. These limitations restrict the application of pure ViT models in settings in which access to vast amounts of data is limited. As a result, small, subtle details often go unnoticed, whereas explanations derived from the model's attention patterns may not correspond to clinically relevant retinal features. RobustDRNet combines the important advantages of CNNs and ViTs to create a model that excels in detecting subtle lesions within an entire retinal image. The fusion of these models results in substantial benefits to the accuracy and a clear understanding of the predictions. While previous studies such as (Hayati et al., 2023) achieved impressive performance (97.6% accuracy) but lacked multi-stage explainability, RobustDRNet not only identified retinal disease stages with high accuracy (88.4%) and a strong AUC (0.967) but also provided five quantitative explanations for its decision-making.

Selvaraju et al. (2017) investigated explainability using Grad-CAM++ and TCAV in individual CNNs. However, RobustDRNet is the first system to integrate seven different XAI methods and evaluate them against both lesion-level ground-truth labels and clinically relevant con-

cepts. RobustDRNet is a unique study owing to its extensive evaluation methodology.

Prior methods either did not leverage interpretability in their evaluation (Bhoopal et al., 2024; Kim et al., 2018), or the datasets used did not correspond to real-world screening scenarios. RobustDRNet's robust evaluation approach is uniquely suited for all DR classes because of class-specific metrics, Cohen's kappa scores, and SHAP values. In addition, TCAV scores achieved an exact match with the conventional terms used in clinical practice, indicating that RobustDRNet provides meaningful explanations that are comprehensible to expert ophthalmologists.

RobustDRNet combines exceptional accuracy, explainability, and robust validation to provide a reliable and clinically usable method to evaluate the presence of DR. When compared to previous CNN-only or ViT-only methods, RobustDRNet stands out for its superior performance and compliance with the rapidly evolving demand for responsible AI approaches in the field of ophthalmology.

Note on comparability across studies. We acknowledge that direct numerical comparison across published methods can be confounded by differences in datasets, preprocessing, train–test splits, and evaluation protocols. Therefore, alongside reporting controlled comparisons among our constituent backbones, fusion baseline, and RobustDRNet under the same experimental setting, we provide a structured literature-based comparison to representative recent hybrid and transformer-based DR grading methods, e.g., DR-CCTNet (Khan et al., 2023), DrViT (Wu et al., 2021), and DR-Trans (Mohan et al., 2022), to contextualize the contribution without making unfair cross-protocol claims.

6.6. Limitations and future work

Despite the good performance of RobustDRNet in both tasks, its limitations suggest areas for improvement in future work. The model is developed on the APTOS 2019 dataset and its lesion-level explanations tested on IDRiD. These datasets offer detailed and realistically applied cases, yet do not fully reflect the wide range of anatomies and imaging conditions observed in actual practice worldwide. Variations in imaging devices, wide variability in retinal pigmentation and heterogeneity in image quality may impede successful cross-cultural model deployment. Future work can benefit from the use of varied, extensive and multicenter data to assess how the model adapts to new situations and mitigate any inconsistencies.

Although we used multiple XAI approaches, all explanations are generated after training the model. However, this increases the possibility of an explanatory model failing to match the way a model makes predictions during inference. A new approach that can leverage interpretable architectures during training to guide the model to reason about diagnostic information in a way consistent with clinical understanding may be a valuable advancement in the future.

According to TCAV evidence, clear separation of clinically related elements (such as microaneurysms, hemorrhages and exudates) emerges within the last classification layer but not in earlier embedding layers. The model's hidden representations are not semantically planned until the final classification phase. Introducing feature regularization strategy elicits more transparent intermediate features and encourages earlier instance discrimination (Yamaguchi et al., 2024).

The model's superior performance comes at the price of higher computational overhead because it utilizes an ensemble of ResNet34, ConvNeXt-Tiny and ViT-B/16. Generating rapid predictions on limited devices is a persistent obstacle in scenarios like remote clinics or mobile health networks. Latency reduction methods like knowledge distillation, pruning or low-rank models may make it possible to deploy models that are both lightweight and accurately reflect the explainable insights of their heavier counterparts.

The current method examines single images in isolation from a patient's complete medical record and neglects data such as OCT scans and blood glucose measurements. Linking the model to additional pa-

tient information and image features is likely to improve its prediction and increase its interpretability.

More work is needed to determine how to use and compare existing evaluation criteria when assessing interpretability in a clinical setting. Some measures assess how accurately important characteristics are interpreted, but not that these interpretations prove helpful to doctors. Participation of ophthalmologists in clinical studies will reveal if interpretable explanations improve trust, decrease errors in diagnosis and affect the preferences of patients, clinicians and healthcare organizations.

Despite the challenges, the results of the study clearly show the remarkable potential offered by RobustDRNet. As a result, the analyzed difficulties reveal the many barriers to clinical AI adoption and highlight ways to improve it. Conquering these challenges will empower the framework's advancement and improve its acceptance, comprehensibility, and usability. The advancements made by this approach increase the possibility of effectively using AI in diabetic retinopathy image analysis.

7. Conclusion

Diabetic retinopathy (DR) remains one of the major causes of preventable vision loss, while existing deep learning techniques have poor classification accuracy and lack interpretability while suffering from class imbalance and opaque decision-making. To solve these problems, we propose RobustDRNet, a hybrid ensemble model that combines Convolutional Neural Networks with Vision Transformers based on a dual fusion strategy. This study presents the training and evaluation of the model on the APTOS dataset and performs comprehensive quantitative validation of this explainability suite via a set of gradient-based, perturbation-based, and concept-based techniques, with expert-labeled lesion data from the IDRiD dataset. In addition to improving the diagnostic accuracy at all DR severity levels, RobustDRNet can provide interpretable and clinically aligned explanations that can serve as a practical decision support tool for ophthalmologists. The results are promising, but remain to be validated on larger, multi-center datasets. In future studies, we will improve generalizability through domain adaptation, model disease progression, and examine real-world deployment in a clinical screening environment.

CRedit authorship contribution statement

Pir Bakhsh Khokhar: Conceptualization, Methodology, Software, Data curation, Formal analysis, Investigation, Visualization, Writing – original draft; **Viviana Pentangelo:** Methodology, Validation, Writing – review & editing; **Carmine Gravino:** Resources, Funding acquisition, Supervision, Project administration, Writing – review & editing; **Fabio Palomba:** Conceptualization, Supervision, Writing – review & editing.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

I, Pir Bakhsh Khokhar, hereby confirm that, to the best of my knowledge, there are no financial or personal interests that might produce a conflict of interest that might affect the contents of the paper. The European Union has partially supported this work through the Italian Ministry of University and Research under the Project PNRR "D3-4Health: Digital Driven Diagnostics, Prognostics, and Therapeutics for Sustainable Healthcare. All the conclusions and opinions expressed within the work are personal and based on the author's independent analysis. Moreover, I assure you that any materials used from other sources or from outside this work have been cited to satisfaction and are in line with the requirements of this institution.

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